

Effect of Incident Light Wavelength and Corneal Edema on Light Scattering and Penetration: Laboratory Study of Human Corneas

Donald A. Peyrot, PhD; Florent Aptel, MD; Caroline Crotti, MSc; Florent Deloison, MSc; Simon Lemaire, BIng; Tal Marciano, MSc; Stéphane Bancelin, MSc; Fatima Alahyane, BIng; Laura Kowalczyk, PhD; Michèle Savoldelli, Eng; Jean-Marc Legeais, MD; Karsten Plamann, PhD

ABSTRACT

PURPOSE: The outcome of ultrashort pulse laser surgery of the cornea is strongly influenced by the light scattering properties of the tissue, for which little data are available. The purpose of the present study is to provide quantitative values for light scattering and its relation to the degree of edema.

METHODS: An experimental optical measuring setup based on confocal geometry was used to measure the unscattered and scattered fractions of light transmitted by eye bank corneas presenting various degrees of edema. From these measurements, the effective light penetration depth in the cornea was calculated as a function of wavelength.

RESULTS: Corneal transparency depends on the pathological state of the cornea and on wavelength. It may be predicted as a function of corneal thickness, ie, the degree of edema. In healthy and edematous cornea, the percentage of scattered light decreases with increasing wavelength. The total penetration depths at the wavelengths of ~1050 nm (which is used in typical clinical systems) and 1650 nm (which is recommended for future devices) are comparable; however, the former is limited by scattering, which degrades the laser beam quality, whereas the latter is only limited by optical absorption, which may be compensated for.

CONCLUSIONS: The use of longer wavelengths should help improve the surgical outcome in ultrashort pulse laser surgery of the cornea when working on pathological tissue. A wavelength of approximately 1650 nm appears to be a good compromise, as it allows for reduced light scattering while keeping optical absorption reasonably low. [*J Refract Surg.* 2010;26(10):786-795.]
doi:10.3928/1081597X-20100921-04

During the past decade, considerable progress has been made in ultrashort pulse laser surgery of the anterior segment of the eye.¹⁻⁸ Ultrashort pulse laser systems are now routinely used for LASIK with a femtosecond laser used to create the corneal flap. Femtosecond LASIK is performed on clear cornea, which permits tight focusing of the surgical laser beam. However, when current systems are used for corneal grafting, which requires surgery in the depth of pathological tissue, the light scattering processes deteriorate the laser beam quality, limit the effective penetration depth of the laser beam, and distort and enlarge the focal spot,^{8,9} which may lead to irregular or failed cuts and unwanted side effects. As most indications for corneal grafting are related to pathologies that cause edema, it is of high interest to elucidate the light scattering process involved, to quantify its contribution, and to identify potential solutions to the problem.

Light propagating through the cornea experiences optical absorption and scattering. For all practical purposes discussed here, the absorption spectrum of the cornea may be considered identical to the one of water, for which precise measurements are readily available. However, the influence of light scattering processes is more difficult to quantify. Figure 1 (left) shows a Siemens star projected through a typical edematous cornea. The high spatial frequencies close to the center are blurred and the transmitted light appears slightly red, which indicates a wavelength dependence of the underlying light

From Laboratoire d'Optique Appliquée, ENSTA ParisTech – École Polytechnique, Palaiseau (Peyrot, Crotti, Deloison, Lemaire, Marciano, Bancelin, Alahyane, Kowalczyk, Plamann); Service d'Ophtalmologie, Hôpital Édouard Herriot, Lyon (Aptel); and Laboratoire Biotechnologie et Œil (EA 4063), Université Paris Descartes (Paris V) – Hôpital Hôtel-Dieu, Paris (Savoldelli, Legeais), France.

The authors acknowledge financial support by the French National Research Agency (ANR) through its TecSan program (project GRECO, ANR-06-TecSan-025). The PhD scholarships for C. Crotti, F. Deloison, and T. Marciano were granted by the French Direction Générale de l'Armement, AXA Research Fund, and CNRS, respectively. The authors thank the Banque Française des Yeux for providing the human cornea samples.

Correspondence: Karsten Plamann, Laboratoire d'Optique Appliquée, ENSTA ParisTech – École Polytechnique – CNRS UMR 7639, Chemin de la Hunière, 91761, Palaiseau cedex, France. E-mail: Karsten.Plamann@ensta-paristech.fr

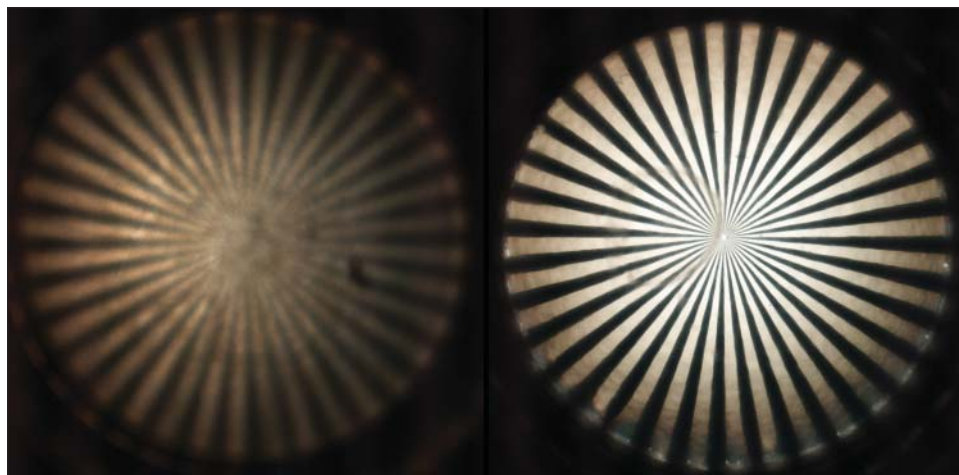


Figure 1. Siemens star viewed through an edematous eye bank cornea (left) and through a cornea having regained its transparency during 4 days of treatment in a Dextran500 deswelling solution (right).

scattering processes. On the right, a cornea is shown that had undergone an osmotic deswelling process, re-establishing a condition close to the one of a typical healthy cornea. The high spatial frequencies are resolved and no appreciable modification of the transmitted light spectrum is apparent.

Several theories have been put forward to explain the optics of healthy and edematous cornea,^{8,10-22} and some details still need clarification. It is generally agreed that corneal transparency is closely linked to the spatial organization of the collagen fibrils in the stroma, which occupy most of the stromal volume. Two light scattering processes are generally identified. As the collagen fibrils in the stroma act as narrow elongated scatterers with diameters much smaller than the wavelength and a refractive index different to the one of the interstitial matrix, they give rise to the so-called Rayleigh scattering. This leads to a wavelength dependency of the scattering cross-section of an individual fibril, which is proportional to λ^{-3} , λ being the wavelength of light.^{12,23-25} The difference in the exponent of the wavelength behavior to the more familiar Rayleigh scattering in the atmosphere (which is proportional to λ^{-4} and explains the blue color of the sky) may be explained as follows: within the spectral range considered here, corneal collagen fibrils act as “one-dimensional” scatterers as opposed to the “zero-dimensional” small particles present in the atmosphere, removing one dimension from the scattering process. As Maurice¹² pointed out in his first study on corneal transparency, if the fibrils act as independent scatterers, even healthy cornea would be opaque. It is the relatively regular arrangement of the fibrils in healthy cornea that results in an effective refractive index sufficiently homogeneous to ensure transparency. Perturbations of the regular structure by edema or other pathologies cause increased scattering, resulting in a reduction of the corneal transparency (from a theoretical point of view, the angular scattering

pattern is proportional to the Fourier transform of the autocorrelation function of the fibril distribution⁸).

The second phenomenon is caused by structures that have dimensions comparable to the wavelength. This light scattering regime is commonly called “Mie” scattering, although strictly speaking, the Mie theory only holds for perfectly spherical particles.

The keratocytes present in the cornea may be responsible for weak light scattering present even in healthy cornea. Their contribution increases with their transformation into myofibroblasts following wound-healing reactions.²⁶ In strongly edematous cornea, “lakes”¹³ of interstitial liquid devoid of collagen fibrils are formed between the lamella causing variations in the average refractive index at micrometric scales giving rise to Mie scattering. Whatever the exact geometry of the scattering particle, no strong wavelength dependence is expected for Mie scattering. However, as this process occurs in addition to the aforementioned Rayleigh scattering, in the presence of strong Mie scattering, a slightly weaker overall wavelength dependence may be observed. Behaviors proportional to λ^{-2} have been reported in the literature for highly edematous cornea,¹⁹ which we attribute to the combination of Rayleigh and Mie scattering processes.

Early experimental studies on light scattering in human corneas were performed by Boettner and Wolter²⁷ who used nine freshly enucleated human eyes and compared “total” transmittance to the unscattered component of the transmitted light in view of quantifying the light scattering in cornea as a function of age. A summary of subsequent experimental work can be found in the research note by van den Berg and Tran.²⁸ More recent studies have focused on light scattering induced by keratectomy^{29,30} and abnormal cellular-based reflections from stromal keratocytes.³¹

The objective of the present study is to complete the available data by providing quantitative measurements

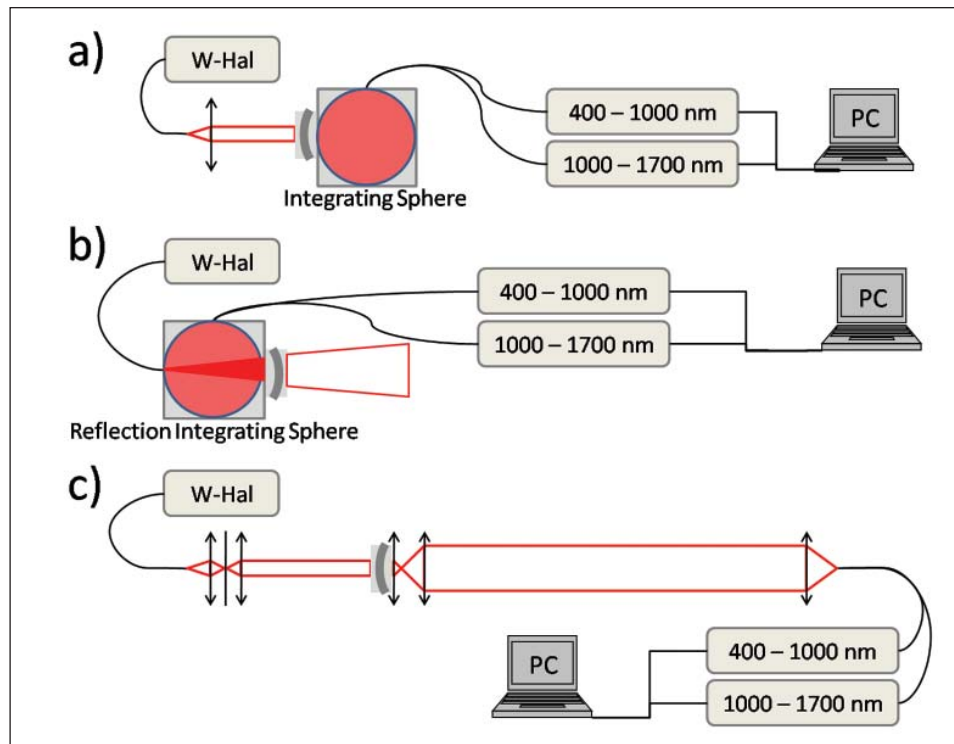


Figure 2. Experimental setups for the measurement of **a)** forward scattered light, **b)** backscattered light, and **c)** directly transmitted light.

of light scattering in healthy and edematous cornea with good spectral resolution over a broad spectrum ranging from the near-infrared (NIR) to the short-wavelength-infrared (SWIR) and the visible spectrum, while maintaining good selectivity by using a very small, solid angle to discriminate the unscattered light from the forward scattered light. These data may then be used to obtain a good estimate for the effective light penetration depth in the cornea as a function of the degree of edema, and to identify new potential wavelengths for the development of new laser sources for femtosecond surgery allowing the influence of scattering to be reduced, as has been done previously for sclera.^{5,6}

MATERIALS AND METHODS

Our method to quantify corneal light scattering is based on the determination of the optical absorption in cornea by the measurement of the total transmission spectrum, ie, the transmitted light regardless of its direction of propagation. This is compared to the measurement of the unscattered part of the transmitted light, which corresponds to the photons experiencing only negligible deviation from their initial trajectory.

The study was conducted according to the tenets of the Declaration of Helsinki and the French legislation for scientific use of human corneas. Fifty human corneas unsuitable for transplantation because of low endothelial cell density were obtained from the Banque Française des Yeux (French Eye Bank), Paris, France. Corneoscleral disks were stored using the organ cul-

ture technique in a CorneaMax medium (Eurobio, Courtabœuf, France) to promote swelling. Storage time ranged from 13 to 15 days.

Corneas are immersed in Hank's balanced salt solution containing Dextran500 (Sigma-Aldrich Chimie S.a.r.l., Lyon, France) in different concentrations. The deswelling rate and the final thickness may be adjusted by varying the concentration.³² Typical Dextran concentrations were 2%, 3%, 4%, 5%, 7%, 10%, and 15% of added weight and the typical deswelling time is less than 5 hours.

Figure 2 shows the experimental setups used for the present study. To suppress surface scattering (as opposed to volume scattering, which is of interest here), the cornea is sandwiched between two uncoated lenses adapted to the corneal curvature, the first lens being plano-concave (LC1906; Thorlabs, Dachau/Munich, Germany) and the second plano-convex (LA1576; Thorlabs) (a similar approach was used by de Brouwere et al³⁰). To ensure index matching, a drop of physiologic serum is inserted between the cornea and the lenses. For normalization purposes, a reference measurement (baseline) is recorded using the identical pair of lenses without the sample, but put into contact with a drop of serum. The open aperture of the lens pair in the lens holder is 8.6 mm.

Two integrating spheres are used to evaluate the absorption spectrum of the cornea. One sphere (FOIS-1; Ocean Optics, Dunedin, Florida) captures the forward scattered light through an aperture of 9.5 mm (Fig 2A)

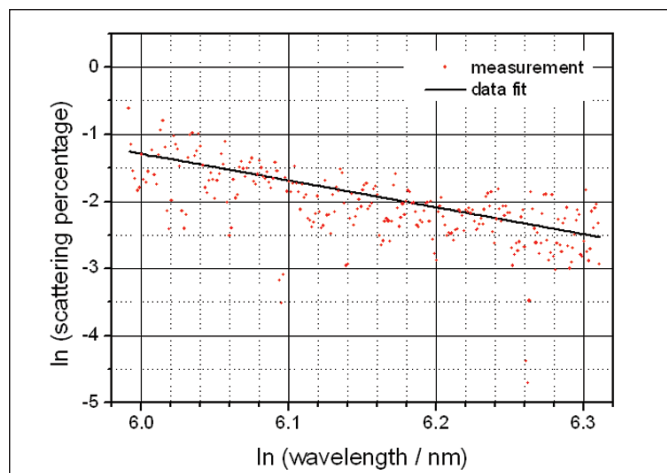


Figure 3. Unscattered transmission measured by the setup shown in Figure 2C through a suspension of 30-nm-thick nanospheres in water plotted as a function of wavelength λ in a double logarithmic chart. The light scattering shows the expected behavior proportional to λ^{-4} typical for Rayleigh scattering by point-like small particles.

and the other (ISP-50-8-R, Ocean Optics) captures the backscattered light (Fig 2B) while the forward transmitted light passes through an aperture of 8 mm. In both cases, shadowing effects due to the difference in diameter between the sphere aperture and the sample holder are taken into account by the baseline measurement.

The measurement of the unscattered transmitted light uses a confocal geometry (Fig 2C). A continuous white light beam from a halogen-tungsten lamp (spectral range: 200 to 2500 nm) is coupled into a multimodal optical fiber while a spatial filter selects a mode covering the entire spectrum without modal superposition. An achromatic lens transforms the filtered mode into a parallel beam with a diameter of 5 mm. The beam passes through the sample placed in front of a telescope. The transmitted light then travels a distance of 750 mm and is then coupled into a split optical fiber with an entry diameter of 600 μm and two output fibers of 200- μm diameter. The solid angle corresponding to the acceptance angle of the confocal setup is $\Omega \approx 3 \cdot \pi \cdot 10^{-8} \text{ sr}$. In the following, by definition, photons that remain within this solid angle will be considered as unscattered. The light is analyzed by two spectrometers sensitive to the visible NIR light (400 to 1000 nm) and NIR-SWIR light (1000 to 1700 nm), respectively.

The telescope is used because the presence or absence of a cornea in the sample holder results in slightly different focal lengths. It is based on a first strongly focusing lens followed by a second lens transforming the light in a parallel beam with a diameter of 8 mm. An adjustment of the position of the second permits compensation for the difference

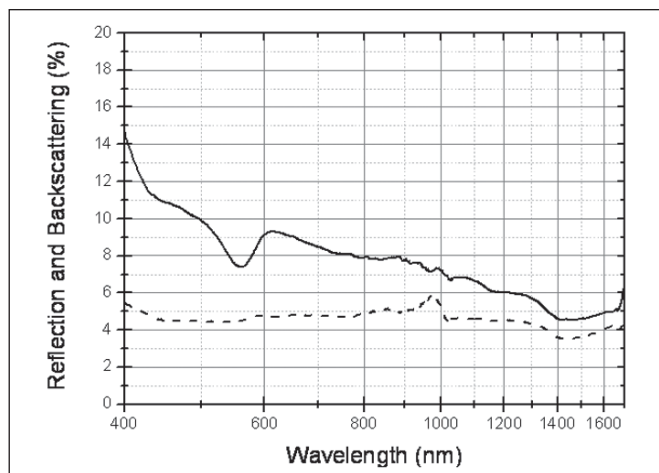


Figure 4. Measurement of the light reflected by an edematous cornea (thickness $\sim 1000 \mu\text{m}$) with and without the sample holder in place (dashed and continuous lines, respectively). The use of the sample holder eliminates backward scattering from the corneal surface and reduces the backward reflection to the Fresnel reflection component.

in focal length. Figure 3 shows a test measurement on a suspension of 30-nm-thick nanospheres in water. A double logarithmic chart shows the expected behavior proportional to λ^{-4} .

A comprehensive presentation of scattering processes²³⁻²⁵ and radiative transfer phenomena³³ can be found in standard textbooks. The parameters that are important for the considerations presented in this article are briefly summarized. When normalized to the incident intensity I_0 , the fractions of the transmitted light T_{FS} , the backward emitted light T_{BS} , and the fraction of light absorbed in the cornea T_A of the light have to add up to unity:

$$(1) \quad T_{BS} + T_{FS} + T_A = 1.$$

In Figure 2, setup (a) measures T_{FS} and setup (b) measures T_{BS} . Figure 4 shows measurements of the backscattered light T_{BS} made with and without the use of the sample holder described above. When the holder is not used, the backscattered light has a significant wavelength dependence. With the holder in place, the backscattered light is almost constant at a value close to 4% over the entire spectral range, which corresponds to the Fresnel reflection for the uncoated first lens used in the sample holder. We may therefore conclude that the sample holder removes the surface backscattering effect. Only a small fraction of the incoming light is backscattered by the volume of the cornea, which is below the sensitivity of our experimental setup.

The transmitted light consists of the unscattered light T_D —which can be measured by setup (c) depicted in Figure 2—and the fraction T_S of the incident light,

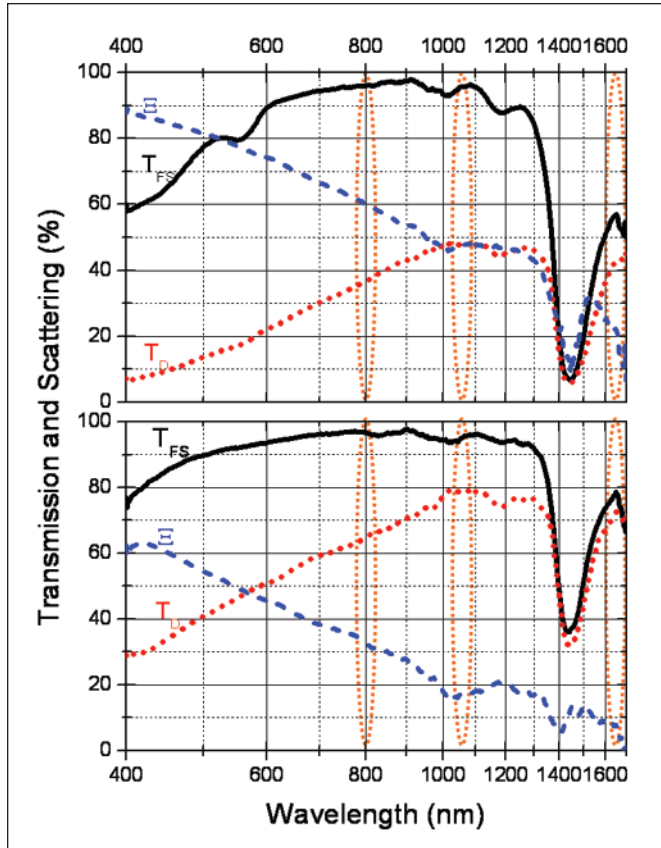


Figure 5. Forward scattered light T_{FS} , unscattered light T_D , and percentage of scattered light for an approximately 1000- μm -thick edematous human cornea (top) and for a 450- μm -thick human cornea (bottom).

which is transmitted by the cornea, but scattered away from the optical axis:

$$(2) \quad T_{FS} = T_D + T_S.$$

The ratio Ξ of the light that is scattered away from the optical axis may be calculated by normalizing T_D by the transmitted light T_{FS} :

$$(3) \quad \Xi = \frac{T_{FS} - T_D}{T_{FS}} = 1 - \frac{T_D}{T_{FS}}.$$

The penetration depth of the light in the tissue may be estimated from the measurements on the scattering spectra. The penetration depth δ_A associated with optical absorption is defined as the distance L in the medium where absorption has reduced the light intensity to a fraction of $1/e$ of its initial value. We may obtain the tissue absorption by Eq.(1). All spectra are normalized with respect to a baseline measurement taken with the sample holder in place, which implicitly takes into account the light reflected backward by Fresnel reflection,

thereby removing T_{BS} from Eq.(1). The transmitted light T_{FS} may be rewritten in terms of penetration depth:

$$(4) \quad T_{FS} = 1 - T_A = e^{-L/\delta_A}.$$

Likewise it is possible to define the effective penetration depth δ_E describing the combined effect of absorption and scattering. We define the effective penetration depth as the distance in the medium where a fraction $1/e$ of the light intensity is not absorbed and not scattered outside of a specific solid angle. This term may be evaluated by rewriting Eq.(2):

$$(5) \quad T_D = e^{-L/\delta_E} = e^{-L/\delta_A} \cdot e^{-L/\delta_S}$$

which permits the definition of the scattering penetration depth δ_S . If the scattering particles are identical and act independently, the scattering penetration depth may be calculated as

$$(6) \quad \delta_S = 1/(\eta \sigma),$$

where η is the density of the scattering particles and σ is their scattering cross-section. This relationship is discussed further below. From Eqs.(4) and (5), δ_E , δ_A , and δ_S may be calculated as

$$(7) \quad \delta_A = -L/\ln(T_{FS}),$$

$$(8) \quad \delta_E = -L/\ln(T_D), \text{ and}$$

$$(9) \quad \delta_S = \frac{-L}{\ln\left(\frac{T_D}{T_{FS}}\right)}.$$

RESULTS

Experimental spectra were measured on 50 eye bank corneas presenting various thicknesses associated to various degrees of edema. Figure 5 shows the typical result obtained for a cornea before and after immersion in the deswelling solution. The total transmitted light spectrum agrees well with earlier data obtained by Boettner and Wolter.²⁷ In untreated cornea, the presence of two absorbance peaks of approximately 400 nm and 550 nm may be attributed to the biological marker “phenol red,” which is present in the storage solution.³⁴⁻³⁷ Both spectra are comparable when taking into account the different thicknesses except for the presence or absence of these absorption bands. Figure 5 also shows the scattering spectra calculated using Eqs.(2) and (3), ie, the percentage of light scattered out of the solid angle defined by the confocal measuring setup. In both cases, the light scattering process shows

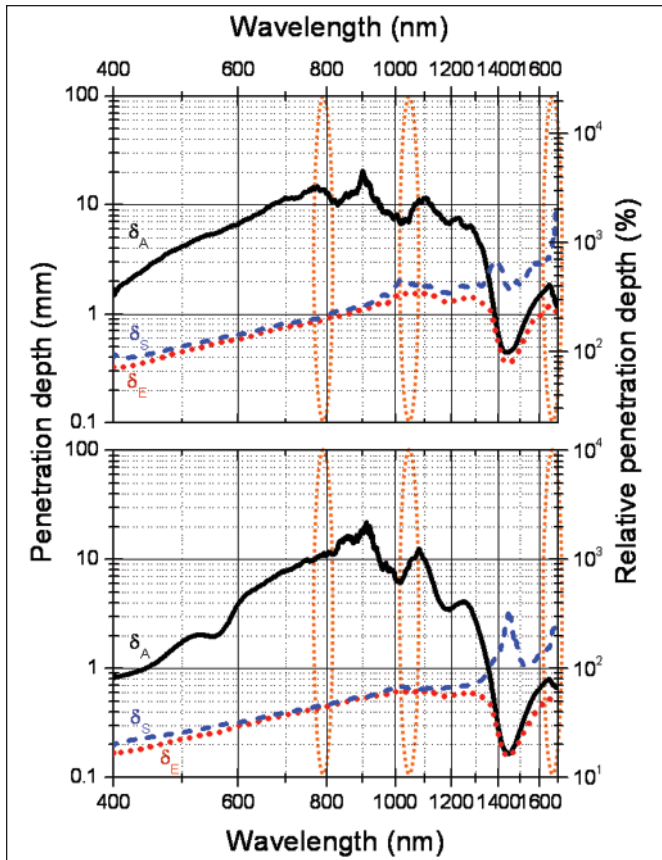


Figure 6. Penetration depth and relative penetration depths for an approximately 1000- μm -thick edematous human cornea (bottom) and for a 450- μm -thick human cornea (top), with the absorption and scattering components, and the resulting effective penetration depth.

a strong wavelength dependence. In the deswollen cornea, the scattering ratio decreases rapidly until 1000 nm and more slowly above this value. The untreated cornea shows stronger scattering value at short wavelength and a less rapid decrease of the scattering of the spectrum. At the wavelength of typical clinical lasers (~ 1050 nm), scattering remains strong. A window of relative optical transparency centered at 1650 nm may be identified where absorption is moderate and scattering is low. In this wavelength range, similar values are measured for the penetration depth as at 1050 nm for the untreated cornea. The apparent minimum in the scattering spectrum close to 1450 nm for healthy cornea is an artifact: in this wavelength range, the strong absorption peak of water dominates the propagation process. Because of the error propagation, small variations in the measured transmission data result in strong errors in the scattering spectrum and the absolute values for the scattering spectra are not significant. This is without consequence for the following discussion in terms of penetration depth.

Although our results confirm that the light scattering

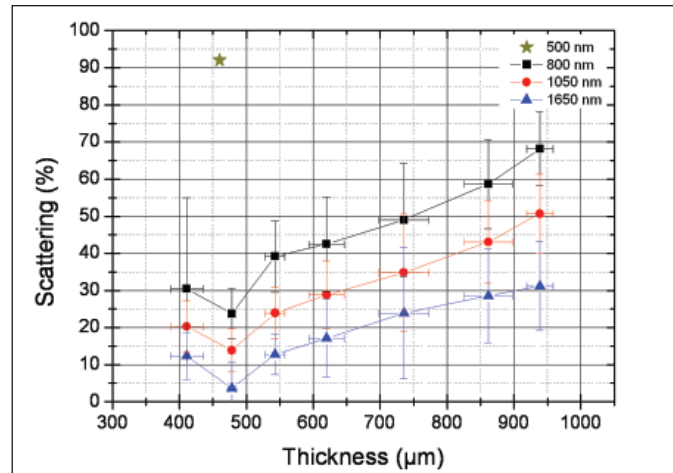


Figure 7. Percentage of scattered light as a function of corneal thickness scattering at 800 nm (black squares), 1050 nm (red dots), and 1650 nm (blue triangles). Error bars indicate standard deviation.

process in human cornea is strongly wavelength dependent (as is the case for most biological tissues³⁸), for practical purposes we may translate the results shown in Figure 5 into penetration depths of the surgical laser beam.

Figure 6 shows the same data presented in Figure 5 but represented in terms of penetration depths as absolute values and values normalized by the corneal thickness. The results obtained for cornea before and after the deswelling treatment are in fact similar in shape apart from the aforementioned artifacts; however, the relative penetration depths with respect to the thickness of the cornea vary strongly.

DISCUSSION

Figures 5 and 6 show that in the visible and NIR spectral range, corneal transparency is limited by light scattering whereas the contribution of optical absorption is negligible. Light scattering is increased by a factor of up to three in the presence of edema. The effective penetration depths (δ_E) are in fact comparable in healthy and edematous cornea. However, when the penetration depth is normalized with respect to the sample thickness, the value is much lower for edematous cornea. This observation may appear unsurprising at first, but its explanation is not entirely trivial: the excess liquid reduces the fibril density per unit volume, which should increase the absolute penetration depth. However this “dilution” effect is compensated by an increased collective scattering cross-section of the fibrils due to the perturbation of their regular arrangement.

Within the window of optical transmission centered at 1.65 μm , the curves for total and unscattered transmission are close: the transmitted light is largely unaffected by scattering. Here, the penetration depth is only limited by the relatively weak optical absorption.

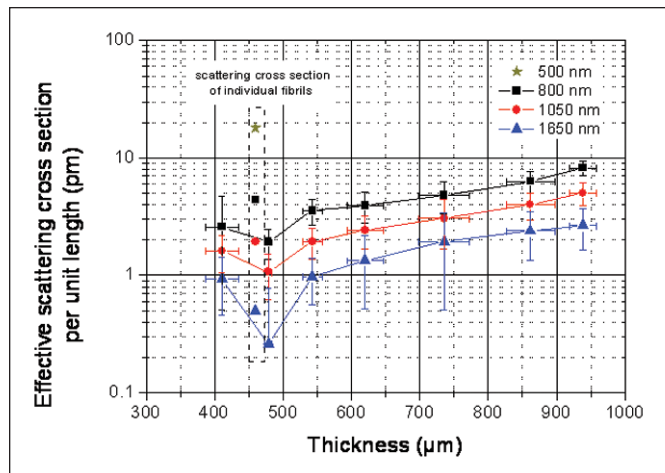


Figure 8. Effective scattering cross-section per unit length for individual fibrils as a function of corneal thickness at 800 nm (black squares), 1050 nm (red dots), and 1650 nm (blue triangles). The dashed rectangle contains the theoretical values for fibrils acting as independent scatterers. The star indicates the value for 500 nm used by Maurice.¹⁰ Error bars indicate standard deviation.

The beam quality of surgical lasers using this wavelength range should be good in pathological cornea.

The Table shows that a surgery system working at a wavelength of approximately 1050 nm would be more efficient than one at 800 nm, because absorption and scattering at 1050 nm are lower, resulting in a higher penetration depth. Comparing 1050 and 1650 nm presents an interesting perspective: at 1650 nm, the effective penetration depth of light is comparable to the one at 1050 nm, but the limitation is not due to the same set of parameters. At 1650 nm, the light penetration is only limited by the residual absorption of the medium while at 1050 nm it is limited by light scattering. This point is of high interest because absorption is a linear process, which is easily compensable by an increase of the laser pulse energy. Numerical calculations have shown that under the given circumstances the effect of the heat deposition in the tissue is negligible (Caroline Crotti, MSc, unpublished data, June 2008). In comparison, the compensation of light scattering is not as straightforward. Furthermore, given that the scattering pattern of cornea is strongly forward peaked, the scattered photons are focused at regions close to the focal spot. The light scattering processes likely cause a considerable enlargement and distortion of the point spread function (PSF) of the incident laser pulse, resulting in incomplete cuts and other unwanted effects and thereby strongly compromising the surgical outcome. This cannot be compensated by simply increasing the pulse energy, as this would cause even stronger unwanted secondary effects.

Additional insight may be gained when plotting the data as a function of corneal thickness, which is

TABLE

Wavelength Comparison of the Effective Penetration Depth in the Cornea and Its Absorption and Scattering Components

Wavelength (nm)	Effective Penetration Depth (mm)	Absorption Component (mm)	Scattering Component (mm)
800	0.7 ± 0.2	10.8 ± 0.4	0.7 ± 0.2
1050	1.2 ± 0.4	13 ± 6	1.3 ± 0.4
1650	0.9 ± 0.2	1.5 ± 0.1	2.1 ± 0.9

an indicator for the degree of edema. Figure 7 shows a comparison of the percentage of scattered light as a function of the corneal thickness for different wavelengths: 800 nm (typical laboratory ultrashort pulsed laser based on titanium-doped sapphire crystals), 1050 nm (typical clinical laser), and 1650 nm. The following conclusions can be made:

- At any stromal thickness, the percentage of scattered light decreases strongly with wavelength. At 1650 nm, the scattering may be reduced by a factor of more than two compared to 800 nm and by approximately 50% with respect to 1050 nm. Even at the physiological stromal thickness (~480 μm without epithelium), scattering at 1650 nm remains significantly weaker than at shorter wavelengths.
- The physiological thickness of the healthy cornea corresponds to a minimum in light scattering. Artificial reduction of the thickness below this value also results in increased scattering.

To obtain a metric for the influence of corneal pathologies, we may express our data in terms of scattering cross-section per fibrils. Following the reasoning of Maurice¹⁰ and using his parameters (see also Hart and Farrell¹¹), we first calculated the theoretical cross-sections per unit length for individual fibrils acting as independent scatterers. These values are plotted in the dashed rectangle in Figure 8; the green star shows Maurice's result for a wavelength of 500 nm. We then used Eq.(6) to calculate and plot the effective cross-section per collagen fibril as a function of corneal thickness while taking into account that corneal swelling dilutes the initial density of the fibrils. The following observations can be made:

- As Maurice concluded, corneal transparency cannot be explained by considering the fibrils as independent Rayleigh scatterers. For all wavelengths of interest here, the regular arrangement of the fibrils within the healthy stroma reduces their scattering cross-section by a factor of approximately two.

- Deviations from the physiological thickness in both directions increase the effective scattering cross-section per individual collagen fibril. This phenomenon is particularly pronounced at 1650 nm.
- For strongly edematous cornea, ie, cornea that has increased more than ~50% in thickness, the effective cross-section appears to reach values that are above those for “independent” scattering.

The last point merits a detailed discussion. Similarly to Maurice’s reasoning that the optical transparency of healthy cornea may not be explained by considering the collagen fibrils as independent Rayleigh scatterers, we point out that it also does not explain the opacity of pathological cornea. Additional light scattering mechanisms obviously contribute.

First, in very thin cornea, the repulsive potential between some fibrils breaks down and some fibrils “stick” together, thereby forming stronger individual scatterers. The combined cross-section is higher than the added cross-sections of individual fibrils (the diameter of one-dimensional Rayleigh scatterers appears with the power of four in the relationship for the scattering cross-section^{10,11,23-25}).

Second, strongly edematous corneas as defined above produce additional Mie scattering notably due to the presence of “lakes,” which results in an overestimation of the fibril scattering cross-section. Additional Mie scattering may be caused by alterations in the keratocytes caused by keratectomy, wound-healing processes, or some pathologies.

Third, even in highly edematous cornea, the collagen fibrils do not act as independent scatterers in the sense of the Rayleigh model: they remain within one-half a wavelength of each other and produce additional interaction processes the Rayleigh theory does not take into account.

We now turn our attention back to the more practical aspects of laser surgery. When considering the practical consequences of a wavelength change, two additional phenomena have to be taken into account. Increasing the wavelength also increases the dimensions of the Airy disk, which a priori should require a corresponding increase in pulse energy. For instance, a wavelength change from 1.05 μm to 1.65 μm would require an increase in pulse energy by a factor of approximately 2.5 to obtain the same radiant exposure. This would obviously compromise the advantage of the reduced light scattering, particularly when taking into account that the optical absorption at 1.65 μm has to be compensated for as well. In clinical practice, however, typical PSFs in tissue are not close to the diffraction limit. As the measurements presented above illustrate, even in healthy cornea, a significant amount

of scattering is present, which causes a broadening of the PSF in the volume of the tissue. As we have shown in another study,³⁹ optical aberrations further contribute to the broadening of the PSF. A rough estimate shows that the additional broadening by these processes is typically wider than the PSF itself and therefore partially or entirely masks the wavelength dependence of the diffraction limited PSF. Experimental evidence exists for this: in our previous surgical experiments,⁸ a wavelength change from approximately 1 to 1.65 μm resulted in an increase in incision depth of a factor of more than three for identical pulse energies, which exceeds the improvement predicted by the reduced scattering after the wavelength change. This may be due to a modification of the threshold for optical breakdown or other effects, the discussion of which is beyond the scope of this article.

A practical application of our results also should take into account that the measurements presented here have been performed in the far field, whereas for the surgical intervention the quality of the focal spot is of interest, which is a near field phenomenon. The scattering fibrils are close to the PSF they perturb and the resulting effective PSF cannot simply be calculated by Fourier-transforming the far field result. As has been shown in the even more strongly scattering sclera, the local PSF breaks down into a closely packed speckle pattern.^{5,6} The exact form of this pattern depends on the position in the tissue; however, the average envelope of the pattern corresponds roughly to a broadened PSF, the form of which can be obtained by Fourier transforming the far field distribution. This is not entirely surprising in as much as the difference in refractive index between the fibrils and matrix is <10%. The fibrils may be considered “soft” scatterers, which do not cause strong local maxima in the light intensity. The “lakes” that are present in strong edema as well as normal and altered keratocytes act as “Mie” scatterers in the far field, which cannot immediately be translated into a near field behavior. However, the small local gradient in refractive index is unlikely to create strong local intensity variations leading to unwanted side effects. Our surgical experiments have shown no evidence for the existence of artifactual local “hot spots.” This is particularly true for the longer wavelengths we recommend. When increasing the energy at 1650 nm to compensate for the absorbed energy, the photons reaching the focal region form a well-defined focal spot, which allows precise and reproducible surgical incisions.

Light scattering and light penetration depth in human corneas presenting different degrees of edema have been studied over a wide optical spectrum. At the

wavelengths close to 1 μm , which are typically used by clinical ultrashort pulse lasers, scattering is strong in the presence of edema, which may explain the often unsatisfactory outcome of keratoplasty assisted by ultrashort lasers and other corneal surgical procedures based on these lasers.⁹ Shifting the laser wavelength to the SWIR range while avoiding the water absorption peak centered at 1450 nm allows the influence of light scattering to be considerably reduced. The transparency band centered at 1650 nm should be optimal for surgery of the cornea assisted by ultrashort pulse lasers. In this wavelength range, the effective penetration depth is comparable to the one observed at 1050 nm, but the laser focus remains well defined and the penetration depth is only limited by optical absorption, which is easy to compensate. Given the beam quality of surgical lasers in practical clinical conditions, the additional broadening of the Airy disk due to the increased wavelength is of little influence. A further benefit is that the SWIR region is safe for the eye: no light can reach the retina. Future developments of femtosecond laser sources for corneal surgery should preferentially address this wavelength range.⁴⁰

The expression of our measured data in terms of effective scattering cross-section per collagen fibril illustrates the strong influence of the tissue structure and presence of edema on the corneal scattering properties and may be of use for future theoretical considerations.

AUTHOR CONTRIBUTIONS

Study concept and design (D.A.P., K.P.); data collection (D.A.P., F.L.A., S.L., T.M.); analysis and interpretation of data (D.A.P., F.L.A., C.C., F.D., T.M., S.B., Fa.A., L.K., M.S., J.M.L., K.P.); drafting of the manuscript (D.A.P., K.P.); critical revision of the manuscript (F.L.A., C.C., F.D., S.L., T.M., S.B., Fa.A., L.K., M.S., J.M.L., K.P.); obtained funding (J.M.L., K.P.); administrative, technical, or material support (J.M.L., K.P.); supervision (J.M.L., K.P.)

REFERENCES

1. Soong HK, Katz DG, Farjo AA, Sugar A, Meyer RF. Central lamellar keratoplasty for optical indications. *Cornea*. 1999;18(3):249-256.
2. Stern D, Schoenlein RW, Puliafito CA, Dobi ET, Birngruber R, Fujimoto JG. Corneal ablation by nanosecond, picosecond, and femtosecond lasers at 532 and 625 nm. *Arch Ophthalmol*. 1989;107(4):587-592.
3. Roach WP, Rogers ME, Rockwell BA, Boppart SA, Stein CD, Bramlette CM. Ultrashort laser pulse effects in ocular and related media. *Aviat Space Environ Med*. 1994;65(5 Suppl):100-107.
4. Juhasz T, Kastis GA, Suarez C, Bor Z, Bron WE. Time-resolved observations of shock waves and cavitation bubbles generated by femtosecond laser pulses in corneal tissue and water. *Lasers Surg Med*. 1996;19(1):23-31.
5. Sacks ZS, Kurtz RM, Juhasz T, Mourou GA. High precision subsurface photodisruption in human sclera. *J Biomed Opt*. 2002;7(3):442-450.
6. Sacks ZS, Kurtz RM, Juhasz T, Spooner G, Mourou GA. Subsurface photodisruption in human sclera: wavelength dependence. *Ophthalmic Surg Lasers Imaging*. 2003;34(2):104-113.
7. Soong HK, Malta JB. Femtosecond lasers in ophthalmology. *Am J Ophthalmol*. 2009;147(2):189-197.
8. Plamann K, Aptel F, Arnold CL, Courjaud A, Crotti C, Deloison F, Druon F, Georges P, Hanna M, Legeais JM, Morin F, Mottay É, Nuzzo V, Peyrot DA, Savoldelli M. Ultrashort pulse laser surgery of the cornea and the sclera. *Journal of Optics*. 2010;12(8). doi:10.1088/2040-8978/12/8/084002.
9. Nuzzo V, Plamann K, Savoldelli M, Merano M, Donate D, Albert O, Gardeazabal Rodríguez PF, Mourou G, Legeais JM. In situ monitoring of second-harmonic generation in human corneas to compensate for femtosecond laser pulse attenuation in keratoplasty. *J Biomed Opt*. 2007;12(6):064032.
10. Maurice DM. The structure and transparency of the cornea. *J Physiol*. 1957;136(2):263-286.
11. Hart RW, Farrell RA. Light scattering in the cornea. *J Opt Soc Am*. 1969;59(6):766-774.
12. Maurice DM. The transparency of the corneal stroma. *Vision Res*. 1970;10(1):107-108.
13. Feuk T. On the transparency of the stroma in the mammalian cornea. *IEEE Trans Biomed Eng*. 1970;17(3):186-190.
14. Feuk T. The wavelength dependence of scattered light intensity in rabbit corneas. *IEEE Trans Biomed Eng*. 1971;18(2):92-96.
15. Farrell RA, McCally RL, Tatham PE. Wave-length dependencies of light scattering in normal and cold swollen rabbit corneas and their structural implications. *J Physiol*. 1973;233(3):589-612.
16. Farrell RA, McCally RL. On corneal transparency and its loss with swelling. *J Opt Soc Am*. 1976;66(4):342-345.
17. Hart RW, Farrell RA. On the theory of the spatial organization of macromolecules in connective tissue. *Bull Math Biophys*. 1969;31(4):727-760.
18. Hart RW, Farrell R. Structural theory of the swelling pressure of corneal stroma in saline. *Bull Math Biophys*. 1971;33(2):165-186.
19. Benedek G. Theory of transparency of the eye. *Appl Opt*. 1971;10(3):459-473.
20. Cox JL, Farrell RA, Hart RW, Langham ME. The transparency of the mammalian cornea. *J Physiol*. 1970;210(3):601-616.
21. Farrell RA, McCally RL. In: Albert DM, Jakobiec FA, Azar DT, Gragoudas ES, eds. *Principles and Practice of Ophthalmology*. 2nd ed. Philadelphia, PA: WB Saunders; 2000.
22. Smith TB. Modeling corneal transparency. *American Journal of Physics*. 2007;75(7):588-596.
23. van de Hulst HC. *Light Scattering by Small Particles*. New York, NY: New Dover Publications Inc; 1984.
24. Kokhanovsky AA. *Light Scattering Media Optics*. 3rd ed. Berlin, German: Springer; 2004.
25. Bohren CF, Huffman DR. *Absorption and Scattering of Light by Small Particles*. Berlin, Germany: Wiley VCH Verlag; 2004.
26. Jester JV, Petroll WM, Cavanagh HD. Corneal stromal wound healing in refractive surgery: the role of myofibroblasts. *Prog Ret Eye Res*. 1999;18(3):311-356.
27. Boettner EA, Wolter JR. Transmission of the ocular media. *Invest Ophthalmol Vis Sci*. 1962;1:776-783.
28. van den Berg TJ, Tan KE. Light transmittance of the human cornea from 320 to 700 nm for different ages. *Vision Res*. 1994;34(11):1453-1456.

29. Del Val JA, Barrero S, Yáez B, Merayo J, Aparicio JA, González VR, Pastor JC, Mar S. Experimental measurement of corneal haze after excimer laser keratectomy. *Appl Opt*. 2001;40(10):1727-1734.
30. de Brouwere D, Ginis H, Kymionis G, Naoumidi I, Pallikaris I. Forward scattering properties of corneal haze. *Optom Vis Sci*. 2008;85(9):843-848.
31. Møller-Pedersen T. Keratocyte reflectivity and corneal haze. *Exp Eye Res*. 2004;78(3):553-560.
32. Niemz MH. *Laser-Tissue Interactions: Fundamentals and Applications*. 3rd ed. Berlin, Germany: Springer-Verlag; 2003.
33. Wolf AH, Welge-Lüssen UC, Priglinger S, Kook D, Grueterich M, Hartmann K, Kampik A, Neubauer AS. Optimizing the deswelling process of organ-cultured corneas. *Cornea*. 2009;28(5):524-529.
34. Pels L. Organ culture: the method of choice for preservation of human donor corneas. *Br J Ophthalmol*. 1997;81(7):523-525.
35. EEBA European Eye Bank Directory. 15th ed. Amsterdam, The Netherlands; 2007.
36. Pels E, Beele H, Claerhout I. Eye bank issues: II. Preservation techniques: warm versus cold storage. *Int Ophthalmol*. 2008;28(3):155-163.
37. Jeng BH. Preserving the cornea: corneal storage media. *Curr Opin Ophthalmol*. 2006;17(4):332-337.
38. Tuchin VV. *Tissue Optics: Light Scattering Methods and Instruments for Medical Diagnostics*. 2nd ed. Bellingham, WA: SPIE Press; 2007.
39. Nuzzo V, Savoldelli M, Legeais J-M, Plamann K. Self-focusing and spherical aberrations in corneal tissue during photodisruption by femtosecond laser. *J Biomed Opt*. 2010;15(3):038003.
40. Morin F, Druon F, Hanna M, Georges P. Microjoule femtosecond fiber laser at 1.6 microm for corneal surgery applications. *Opt Lett*. 2009;34(13):1991-1993.

Copyright of Journal of Refractive Surgery is the property of SLACK Incorporated and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.